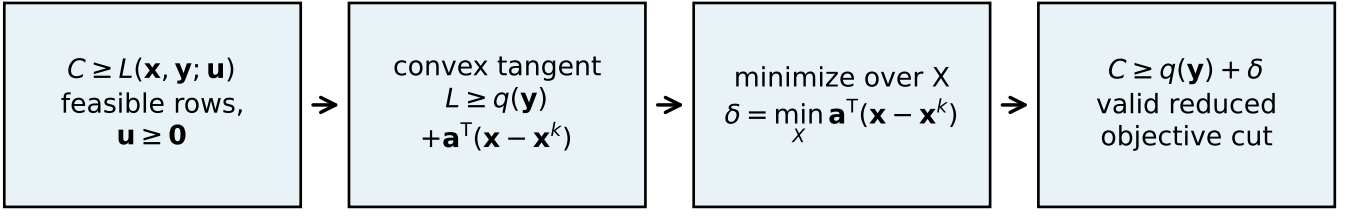
$C \geq L(\mathbf{x}, \mathbf{y}; \mathbf{u})$
feasible rows,
$\mathbf{u} \geq \mathbf{0}$
convex tangent
$L \geq q(\mathbf{y})$
$+\mathbf{a}^{\mathsf{T}}(\mathbf{x} - \mathbf{x}^k)$
minimize over X
$\delta = \min_X \mathbf{a}^{\mathsf{T}}(\mathbf{x} - \mathbf{x}^k)$
$C \geq q(\mathbf{y}) + \delta$
valid reduced
objective cut